

SYMPOSIUM REPORT

Physiological basis of interval training for performance enhancement

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Abstract

Interval training refers to an intermittent style of exercise, in which bouts of more intense effort are interspersed with recovery periods within a given training session. Physiological responses to interval training depend on numerous factors, including the specific nature of the intervention and the initial training state of the individual. Interval training improves performance in part by enhancing the capacity for aerobic energy provision, even in those who are already trained. Two primary mechanisms in this regard are an increased whole-body maximal oxygen uptake and an enhanced capacity for oxidative metabolism in skeletal muscle owing to an increase in mitochondria. In comparison to moderate-intensity continuous exercise, interval training can elicit superior responses when total work is matched, and similar responses despite a reduced training volume and time commitment.

KEYWORDS

exercise, maximal oxygen uptake, mitochondrial biogenesis, skeletal muscle

1 | INTRODUCTION

Interval training refers to an intermittent style of exercise, in which bouts of more intense effort are interspersed with recovery periods within a given training session. It has long been practised by high-level athletes as a strategy to enhance performance, particularly in events that rely heavily on the capacity for aerobic energy provision, such as middle- and long-distance running (Billat, 2001a, b). This capacity is also important for sports characterized by repeated, high-intensity efforts, such as football (soccer), in which aerobic energy has been estimated to account for >90% of total energy provision (Bangsbo, 1994). Research on the practice dates to the middle of the 20th century, but the last two decades have seen a resurgence of scientific inquiry into the responses to interval training that are associated with performance enhancement. The determinants of performance are obviously complex and include a broad range of psychological, technical and tactical characteristics, in addition to physiological attributes (Bangsbo, 2015).

Physiological responses to interval training depend on numerous factors, including the specific nature of the intervention and the initial training state of the individual. With regard to protocol characterization, interval training is infinitely variable, and there is a

lack of standardization, both in the scientific literature and amongst practitioners, regarding the terminology used to describe different approaches. One general classification scheme considers two broad subtypes (MacInnis & Gibala, 2017): high-intensity interval training (HIIT), also referred to as aerobic interval training, and sprint interval training (SIT), also called anaerobic interval training or speed endurance training. HIIT involves vigorous but submaximal efforts that generally elicit $\geq 80\%$ of maximal heart rate, and longer bouts, typically lasting up to ~ 5 min. SIT describes a more intense version of the method that involves 'near-maximal' to 'all-out' efforts, in which the absolute workload exceeds that which would elicit maximal oxygen uptake ($\dot{V}_{O_{2\max}}$) and, typically, shorter bouts lasting ≤ 45 s. This brief review considers the effect of interval training on primary physiological determinants of aerobic energy provision, which are linked to performance (Hostrup & Bangsbo, 2017; Lundby & Jacobs, 2016; Lundby, Montero, & Joyner, 2017).

2 | INTERVAL TRAINING AND MAXIMAL OXYGEN UPTAKE

$\dot{V}_{O_{2\max}}$ establishes the maximal upper limit for aerobic energy provision and is a key determinant of endurance performance (Lundby

et al., 2017). Systematic reviews and meta-analyses have concluded that a wide variety of interval training protocols, lasting from a few weeks to up to several months, increase $\dot{V}_{O_{2\max}}$ (used interchangeably here with peak oxygen uptake) in healthy sedentary and recreationally active individuals (Bacon, Carter, Ogle, & Joyner, 2013). For a given 'dose' or fixed amount of exercise (e.g. when total energy expenditure is matched between protocols), the increase in $\dot{V}_{O_{2\max}}$ after interval training appears to be greater than for moderate-intensity continuous training (MICT; Milanović, Sporiš, & Weston, 2015). There is considerable variability in training responsiveness, but relatively high-volume protocols that incorporate intense intervals appear to elicit the greatest increases in $\dot{V}_{O_{2\max}}$ (Bacon et al., 2013; Lundby et al., 2017). The potential for brief, vigorous intermittent exercise and SIT to increase $\dot{V}_{O_{2\max}}$ in a time-efficient manner (Sloth, Sloth, Overgaard, & Dalgas, 2013; Wen et al., 2019) is also noteworthy from a broader health perspective. This is owing to the fact that the clinical correlate of $\dot{V}_{O_{2\max}}$, cardiorespiratory fitness, is associated with risk for all-cause mortality (Lundby et al., 2017), and given that 'lack of time' is a commonly cited barrier to physical activity. The potential for training-induced improvements is limited in very high-level athletes (Bangsbo, 2015), but there is evidence that $\dot{V}_{O_{2\max}}$ can be increased further in these individuals. A recent study showed that nine sessions of SIT over 3 weeks increased $\dot{V}_{O_{2\max}}$ in elite cyclists (from 73.3 ± 3.6 to 75.5 ± 2.7 ml kg⁻¹ min⁻¹), whereas an effort-matched HIIT protocol did not (Rønnestad, Hansen, Nygaard, & Lundby, 2020).

Numerous physiological factors interact in a complex manner to regulate $\dot{V}_{O_{2\max}}$. As summarized by the Fick equation, these can be grouped broadly into the 'central' and 'peripheral' determinants of maximal cardiac output and the maximal arteriovenous difference in blood oxygen content, respectively. Training-induced increases in $\dot{V}_{O_{2\max}}$ are believed to be facilitated largely by expansion of red blood cell volume and an associated increase in stroke volume, which also adapts independent of changes in red blood cell volume (Lundby et al., 2017). Warburton et al. (2004) reported that 12 weeks of work-matched HIIT and MICT in untrained individuals induced similar improvements in $\dot{V}_{O_{2\max}}$ and peak stroke volume, which was determined using radionuclide ventriculography. Blood volume also increased to a similar extent in both groups, with hypervolaemia accounting for ~47% of the change in $\dot{V}_{O_{2\max}}$. Helgerud et al. (2007) examined the effect of four different training protocols, performed three times per week for 8 weeks, in groups of trained men who had a mean $\dot{V}_{O_{2\max}}$ of ~55–60 ml kg⁻¹ min⁻¹. Two protocols involved HIIT interventions that elicited >90% of maximal heart rate. Both protocols increased $\dot{V}_{O_{2\max}}$ and stroke volume, which was determined based on the non-invasive measurement of cardiac output using a single-breath acetylene uptake method. In contrast, there was no change in $\dot{V}_{O_{2\max}}$ or stroke volume after two matched-work continuous training protocols, which were performed at intensities of ~70 or 85% of maximal heart rate. Helgerud et al. (2007) found no training-induced change in red cell volume or blood volume, the latter of which was measured in only one of the continuous and one of the interval groups. Recent work has also supported the view that HIIT-induced increases

New Findings

- **What is the topic of this review?**

This review considers the physiological basis of interval training for performance enhancement, with an emphasis on the capacity for aerobic energy provision.

- **What advances does it highlight?**

It highlights advances regarding the effect of interval training on primary physiological determinants of aerobic energy provision, which are associated with performance.

in $\dot{V}_{O_{2\max}}$ are associated with an enhanced cardiac output (Astorino et al., 2017). Astorino et al. (2017) reported increases in stroke volume and maximal cardiac output, determined using an impedance cardiograph device, after 20 sessions of interval training performed over 6 weeks, in active young men and women.

3 | SKELETAL MUSCLE RESPONSES TO INTERVAL TRAINING

Studies of various interval training interventions lasting ≤6 weeks have generally reported no change in maximal cardiac output or haematological factors, despite an increased $\dot{V}_{O_{2\max}}$ (Jacobs et al., 2013; Macpherson, Hazell, Oliver, Paterson, & Lemon, 2011; Raleigh et al., 2018). Although not measured directly in these studies, it was suggested, based on the Fick principle, that enhanced peripheral oxygen extraction by skeletal muscle might facilitate the early increase in $\dot{V}_{O_{2\max}}$ seen after a few weeks of interval training. The primary skeletal muscle responses that potentially contribute to training-induced changes in oxygen extraction are capillarization and enhanced mitochondrial volume (Lundby et al., 2017). Short-term interval training can induce vascular remodelling, as evidenced by increased skeletal muscle microvascular and capillary density (Cocks et al., 2013; Scribbans et al., 2014), in previously sedentary and active individuals, which could potentially improve local blood perfusion. Interval training is also a potent stimulus for mitochondrial biogenesis and rapidly increases biomarkers of mitochondrial content and respiration (Jacobs et al., 2013). The capacity for skeletal muscle oxidative respiration exceeds the capacity for systemic oxygen transport, however, and training-induced increases in capillarization and mitochondrial volume density are typically larger than those for $\dot{V}_{O_{2\max}}$ (Lundby & Jacobs, 2016; Lundby et al., 2017).

The functional significance of training-induced increases in mitochondrial volume density is likely to be an enhanced capacity for substrate oxidation, including a greater utilization of lipid and reduced carbohydrate catabolism at a given absolute workload (Lundby & Jacobs, 2016). The potency of SIT, in particular, for eliciting mitochondrial remodelling is evidenced by a study that showed as little as three, 20 s 'all-out' sprints, set within a 10 min session

that otherwise involved low-intensity cycling, increased mitochondrial content to a similar extent as a 50 min MICT protocol, when performed three times per week for 12 weeks in previously sedentary individuals (Gillen et al., 2016). An acute bout of SIT has also been shown to elicit greater activation of signalling cascades associated with mitochondrial biogenesis in comparison to a bout of continuous exercise despite reduced time commitment (Granata, Oliveira, Little, Renner, & Bishop, 2017), even in trained individuals (Skovgaard, Brandt, Pilegaard, & Bangsbo, 2016). The precise regulation of skeletal muscle mitochondrial responses to training, including the role of intensity, duration and volume, remains to be elucidated fully. It has been proposed that intensity is a key determinant of training-induced changes in mitochondrial respiration, whereas volume might be more important for changes in mitochondrial content (Granata, Jamnick, & Bishop, 2018). A recent comprehensive review by Bishop and colleagues (2019) considers current controversies in this field and highlights some important methodological issues that need to be addressed to resolve existing conflicts and fundamental questions.

Limited evidence suggests that interval training is superior to work-matched MICT for increasing mitochondrial content. MacInnis et al. (2017) adopted a single-leg cycling model to facilitate a unique within-subject comparison of mitochondrial responses to a short-term HIIT or MICT protocol matched for total exercise volume and mean training intensity. Participants performed six sessions of training with each leg, which were randomized to perform 30 min of continuous exercise at 50% of single-leg peak workload, or four, 5 min intervals at 65% of peak workload interspersed by 2 min of active recovery. Both protocols increased mitochondrial content, as indicated by the maximal activity of citrate synthase, but the increase was greater in the leg that performed HIIT compared with MICT. These data suggest that exercise intensity and/or the pattern of contraction is an important determinant of exercise-induced skeletal muscle remodelling in humans. Metabolic fluctuations caused by the succession of on- and off-transients during intermittent exercise could trigger divergent responses in comparison to continuous exercise (Combes et al., 2015). In this regard, intermittent in comparison to work-matched continuous exercise was reported to induce a greater activation of signalling pathways linked to the transcriptional coactivator peroxisome proliferator-activated receptor γ coactivator-1 α (PGC-1 α), which is an important regulator of mitochondrial biogenesis. Higher-intensity, intermittent exercise has also been shown to induce greater activation of AMP-activated protein kinase in type II fibres of human skeletal muscle, in comparison to continuous exercise (Kristensen et al., 2015), suggesting the potential for fibre-type-specific responses to the two forms of training.

Particularly in more highly trained individuals, interval training can elicit improvements in performance without measurable changes in markers of mitochondrial content (Bangsbo, 2015). For example, Skovgaard et al. (2017) found that runners who trained using repeated 30 s sprints enhanced short-term exercise capacity and 10 km performance, but the maximal activity of citrate synthase was unchanged. Rather, the improved performance was associated with increased expression of muscle proteins related to sodium/potassium transport and calcium reuptake. Another study (Skovgaard et al.,

2018) suggested the potential for improved running economy after intense intermittent training, which might be related to changes in the expression of proteins linked to energy-consuming processes, mainly in type I muscle fibres. A recent comprehensive review by Hostrup and Bangsbo (2017) considers the importance of changes in muscle ion handling, including the regulation of calcium, hydrogen, lactate and potassium, after intense interval training.

4 | CONCLUSION

Interval training can elicit physiological responses that enhance the capacity for aerobic energy provision and performance, beyond that achieved through moderate-intensity continuous training alone. The precise role of the training 'impulse', the time course for changes and the inter-individual variation in responsiveness, including potential sex-based differences and the influence of training status, remain fruitful areas of investigation.

COMPETING INTERESTS

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